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$$\lim_{x\to\infty} \frac{a_m x^m + \cdots + a_0}{a_n x^n + \cdots + a_n} = \begin{cases} \infty & \text{if } m > n \\ \frac{a_m}{a_n} & \text{if } m = n \end{cases}$$

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 \Longrightarrow \lim_{x\to 0} k \frac{\sin(kx)}{(kx)} = k$$

$$\lim_{x\to\infty} \frac{3x^3+5x}{x^3+1} = 3$$

$$= \lim_{x\to\infty} \frac{\frac{3x^3}{x^3} + \frac{5x}{x^3}}{\frac{x^3}{x^3} + \frac{1}{x^3}} = \lim_{x\to\infty} \frac{3 + \frac{5}{x^2}}{1 + \frac{1}{x^3}} = 3$$

$$\lim_{x\to\infty} \frac{3x^3+5x}{x^2+1} = \infty$$

$$\Downarrow$$

$$= \lim_{x} \frac{3x^3 + 5x}{\ldots} \cdot \frac{3 + \frac{5}{x^2}}{\ldots} = 3 = \infty$$

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Rules of limits

① $\lim_{x \to 0} \frac{\sin x}{x} = 1$, $\lim_{x \to 0} \frac{\sin kx}{x} = k$

② $\lim_{x \to 0} \frac{\tan x}{x} = 1$ $\lim_{x \to 0} \frac{\tan kx}{x} = k$

③ $\lim_{x \to a} \frac{x^n - a^n}{x - a} = n a^{n-1}$

④ $\lim_{x \to a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}$

⑤ $\lim_{n \to 0} \left(1 + \frac{x}{n}\right)^n = e^x$, $\lim_{n \to \infty} (1 + nx)^{\frac{1}{n}} = e^x$

⑥ $\lim_{x \to \infty} \frac{1}{x} = 0$ $\lim_{x \to -\infty} \frac{1}{x} = 0$

$$\lim_{x \to a} \frac{a_n x^n + \cdots + a_0}{a_m x^m + \cdots + a_0} = \begin{cases} 0 & m < n \\ \infty & m > n \\ \frac{a_n}{a_m} & m = n \end{cases}$$

$x \to 0$ 262 554

$= K$

$$\lim_{x\to\infty} \frac{3x^3+5x}{x^3+1} = 3$$

$$= \lim_{x\to\infty} \frac{\frac{3x^3}{x^3}+\frac{5x}{x^3}}{\frac{x^3}{x^3}+\frac{1}{x^3}} = \lim_{x\to\infty} \frac{3+\frac{5}{x^2}}{1+\frac{1}{x^3}} = 3$$

$$\lim_{x\to\infty} \frac{3x^3+5x}{x^2+1} = \infty$$

$$\Rightarrow = \lim_{x\to\infty} \frac{\frac{3x^3}{x^2}+\frac{5x}{x^3}}{\frac{x^2}{x^2}+\frac{1}{x^2}} = \lim_{x\to\infty} \frac{3+\frac{5}{x^2}}{\frac{1}{x}+\frac{1}{x^2}} = \frac{3}{0} = \infty$$

① $\lim_{x \to 3} \frac{x^3 - 27}{x - 3} = \frac{0}{0}$

$= \lim_{x \to 3} \frac{\cancel{(x-3)}(x^2+3x+9)}{\cancel{(x-3)}} = \lim_{x \to 3} x^2+3x+9$

$= 9+9+9 = 27$

② $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4} = \frac{0}{0}$

$= \lim_{x \to 4} \frac{\sqrt{x} - 2}{(\sqrt{x})^2 - (2)^2} = \lim_{x \to 4} \frac{\cancel{(\sqrt{x} - 2)}}{\cancel{(\sqrt{x} - 2)}(\sqrt{x} + 2)}$

$= \lim_{x \to 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{2+2} = \frac{1}{4}$

$\lim_{x \to a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}$

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$\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$

$= \lim_{x \to 4} \frac{x^{\frac{1}{2}} - 4^{\frac{1}{2}}}{x - 4} = \frac{\frac{1}{2}}{1} 4^{\frac{1}{2}-1}$

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① $\lim_{x\to 0} \frac{\sin 5x}{x} = 5$.

② $\lim_{x\to 0} \frac{\tan 3x}{\sin 7x} = \lim_{x\to 0} \frac{\frac{\tan 3x}{x}}{\frac{\sin 7x}{x}}$

$= \frac{\lim_{x\to 0} \frac{\tan 3x}{x}}{\lim_{x\to 0} \frac{\sin 7x}{x}} = \frac{3}{7}$

$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{0}{0}$

$\boxed{1-\cos x = 2\sin^2\frac{x}{2}}$

$\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \cos^2\theta = 1-\sin^2\theta$

$\cos^2\frac{x}{2} = 1-\sin^2\frac{x}{2}$

$\cos x = \cos\left(\frac{x}{2}+\frac{x}{2}\right) = \cos^2\frac{x}{2} - \sin^2\frac{x}{2}$

$= 1-\sin^2\frac{x}{2} - \sin^2\frac{x}{2}$

$= \boxed{1-2\sin^2\frac{x}{2}}$

$1-\cos x = 1-(1-2\sin^2\frac{x}{2})$

Evalu[illegible] following limits

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① $\lim_{x\to 3} \frac{x^3-27}{x-3}$ ② $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$

① $\lim_{x\to 3} \frac{x^3-27}{x-3} = \frac{0}{0}$

$= \lim_{x\to 3} \frac{\cancel{(x-3)}(x^2+3x+9)}{\cancel{(x-3)}} = \lim_{x\to 3} x^2+3x+9$

$= 9+9+9 = 27$

② $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4} = \frac{0}{0}$

$= \lim_{x\to 4} \frac{\sqrt{x}-2}{(\sqrt{x})^2-(2)^2} = \lim_{x\to 4} \frac{\cancel{(\sqrt{x}-2)}}{\cancel{(\sqrt{x}-2)}(\sqrt{x}+2)}$

$= \lim_{x\to 4} \frac{1}{\sqrt{x}+2} = \frac{1}{2+2} = \frac{1}{4}$

حل آخر

$\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$

$= \lim_{x\to 4} \frac{x^{\frac{1}{2}}-4^{\frac{1}{2}}}{x-4} = \frac{\frac{1}{2}}{1} 4^{\frac{1}{2}-1}$

$\lim_{x\to a} \frac{x^n-a^n}{x^m-a^m} = \frac{n}{m} a^{n-m}$

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① $\lim_{x\to 0} \frac{\sin 5x}{x}$ ② $\lim_{x\to 0} \frac{\tan 3x}{\sin 7x}$

③ $\lim_{x\to 0} \frac{1-\cos x}{x^2}$ ④ $\lim_{x\to \frac{\pi}{2}} (\frac{\pi}{2}-x)\sec x$.

Sol

① $\lim_{x\to 0} \frac{\sin 5x}{x} = 5$.

② $\lim_{x\to 0} \frac{\tan 3x}{\sin 7x} = \lim_{x\to 0} \frac{\frac{\tan 3x}{x}}{\frac{\sin 7x}{x}}$

$= \frac{\lim_{x\to 0} \frac{\tan 3x}{x}}{\lim_{x\to 0} \frac{\sin 7x}{x}} = \frac{3}{7}$

$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{0}{0}$

$\boxed{1-\cos x = 2\sin^2\frac{x}{2}}$

$\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \cos^2\theta = 1-\sin^2\theta$

$\cos^2\frac{x}{2} = 1-\sin^2\frac{x}{2}$

$\cos x = \cos(\frac{x}{2}+\frac{x}{2}) = \cos^2\frac{x}{2} - \sin^2\frac{x}{2}$

$= 1-\sin^2\frac{x}{2} - \sin^2\frac{x}{2}$

$= \boxed{1-2\sin^2\frac{x}{2}}$

$1-\cos x = 1-(1-2\sin^2\frac{x}{2})$

$\boxed{1-\cos x = +2\sin^2\frac{x}{2}}$

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Find

① $\lim_{x\to 0} \frac{\sin 5x}{x}$ ② $\lim_{x\to 0} \frac{\tan 3x}{\sin 7x}$

③ $\lim_{x\to 0} \frac{1-\cos x}{x^2}$ ④ $\lim_{x\to \frac{\pi}{2}} \left(\frac{\pi}{2}-x\right)\sec x$.

Sol

① $\lim_{x\to 0} \frac{\sin 5x}{x} = 5$.

② $\lim_{x\to 0} \frac{\tan 3x}{\sin 7x} = \lim_{x\to 0} \frac{\frac{\tan 3x}{x}}{\frac{\sin 7x}{x}}$

$= \frac{\lim_{x\to 0} \frac{\tan 3x}{x}}{\lim_{x\to 0} \frac{\sin 7x}{x}} = \frac{3}{7}$

$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{0}{0}$

$\boxed{1-\cos x = 2\sin^2\frac{x}{2}}$

$\cos x = \cos\left(\frac{x}{2}+\frac{x}{2}\right) = \cos^2 x - \sin^2 x$

$\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \cos^2\theta = 1-\sin^2\theta$

$\cos^2\frac{x}{2} = 1-\sin^2\frac{x}{2}$

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Find

① $\lim_{x \to 0} \frac{\sin 5x}{x}$ ② $\lim_{x \to 0} \frac{\tan 3x}{\sin 7x}$

③ $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$ ④ $\lim_{x \to \frac{\pi}{2}} \left(\frac{\pi}{2} - x\right) \sec x$.

Sol.

① $\lim_{x \to 0} \frac{\sin 5x}{x} = 5$.

② $\lim_{x \to 0} \frac{\tan 3x}{\sin 7x} = \lim_{x \to 0} \frac{\frac{\tan 3x}{x}}{\frac{\sin 7x}{x}}$

$= \frac{\lim_{x \to 0} \frac{\tan 3x}{x}}{\lim_{x \to 0} \frac{\sin 7x}{x}} = \frac{3}{7}$

$\lim_{x \to} 1 - \cos x$ 0

$-1 \le \cos\theta \le 1$

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{0}{0}$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\Rightarrow \cos^2\theta = 1 - \sin^2\theta$$

$$\cos^2\frac{x}{2} = 1 - \sin^2\frac{x}{2}$$

$$\boxed{1 - \cos x = 2\sin^2\frac{x}{2}}$$

$$\cos x = \cos\left(\frac{x}{2} + \frac{x}{2}\right) = \cos^2\frac{x}{2} - \sin^2\frac{x}{2}$$

$$= 1 - \sin^2\frac{x}{2} - \sin^2\frac{x}{2}$$

$$= \boxed{1 - 2\sin^2\frac{x}{2}}$$

$$1 - \cos x = 1 - \left(1 - 2\sin^2\frac{x}{2}\right)$$

$$\boxed{1 - \cos x = +2\sin^2\frac{x}{2}}$$

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \lim_{x\to 0} \frac{2\sin^2\frac{x}{2}}{x^2}$$

$$= 2\lim_{x\to 0}\left(\frac{\sin\frac{x}{2}}{x}\right)^2 = 2\left(\frac{1}{2}\right)^2 = \frac{1}{2}$$

طريقة أخرى

$$\lim_{x\to 0} \frac{1-\cos x}{\ln(\cos x)} \quad \frac{1-\cos^2 x}{}$$

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$$1-\cos x = 1-(1-2\sin^2\tfrac{x}{2})$$

$$\boxed{1-\cos x = +2\sin^2\tfrac{x}{2}}$$

$$\therefore \lim_{x\to 0}\frac{1-\cos x}{x^2} = \lim_{x\to 0}\frac{2\sin^2\frac{x}{2}}{x^2}$$

$$= 2\lim_{x\to 0}\left(\frac{\sin\frac{x}{2}}{x}\right)^2 = 2\left(\frac{1}{2}\right)^2 = \frac{1}{2}$$

طريقة أخرى

$$\lim_{x\to 0}\frac{1-\cos x}{x^2}\cdot\left(\frac{1+\cos x}{1+\cos x}\right) = \lim_{x\to 0}\frac{1-\cos^2 x}{x^2\cdot(1+\cos x)}$$

$$= \lim_{x\to 0}\frac{\sin^2 x}{x^2}\cdot\frac{1}{1+\cos x}$$

$$= \lim_{x\to 0}\frac{\sin^2 x}{x^2}\cdot\lim_{x\to 0}\frac{1}{1+\cos x} = \frac{1}{2}$$

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$$\lim_{x\to 0} \frac{}{x^2}\left(\frac{1+\cos x}{1+\cos x}\right) = \lim_{x\to 0} \frac{1-\cos^2 x}{x^2\cdot(1+\cos x)}$$

$$= \lim_{x\to 0} \frac{\sin^2 x}{x^2} \cdot \frac{1}{1+\cos x}$$

$$= \lim_{x\to 0} \underbrace{\frac{\sin^2 x}{x^2}}_{1} \cdot \lim_{x\to 0} \frac{1}{1+\cos x} = \frac{1}{2}$$

(4) $$\lim_{x\to \frac{\pi}{2}} \left(\frac{\pi}{2}-x\right)\sec x = \lim_{x\to \frac{\pi}{2}} \left(\frac{\pi}{2}-x\right)\cdot\frac{1}{\cos x} \quad 0\cdot\infty$$

let $\frac{\pi}{2}-x = y \Rightarrow x = \frac{\pi}{2}-y$

$$= \lim_{y\to 0} y\cdot\frac{1}{\cos\left(\frac{\pi}{2}-y\right)}$$

$$= \lim_{y\to 0} \frac{y}{\sin y} = \lim_{y\to 0} \frac{1}{\frac{\sin y}{y}} = 1$$

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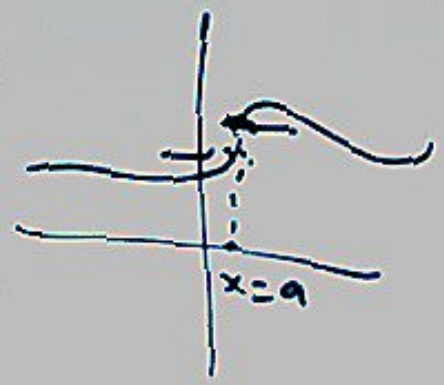
Continuity

الإتصال



Defn

the function $f(x)$ is continuous at $x=a$ if

① $f(a)$ exist

② $\lim_{x \to a} f(x)$ exist.

③ $f(a) = \lim_{x \to a} f(x)$

ex If $f(x) = \begin{cases} \dfrac{\sin 2x}{x} & x \neq 0 \\ 2 & x = 0 \end{cases}$ is $f(x)$ continuous at $x=0$

③ 262 554 $\lim_{x \to a} f(x)$

ex If $f(x) = \begin{cases} \frac{\sin 2x}{x} & x \neq 0 \\ 2 & x = 0 \end{cases}$ is $f(x)$ continuous at $x = 0$

Sol

① $\because f(0) = 2$ ✓

$\lim_{x \to 0} f(x) = \lim_{x \to 0} \frac{\sin 2x}{x} = 2$ ✓

$\therefore f(0) = \lim_{x \to 0} f(x)$

$\therefore f(x)$ is continuous at $x = 0$

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ex

If $f(x) = \begin{cases} \dfrac{x^2-9}{x-3} & x \neq 3 \\ 3 & x = 3 \end{cases}$

is $f(x)$ continuous at $x=3$

Sol.

$\Rightarrow f(3) = 3$

$\lim_{x \to 3} f(x) = \lim_{x \to 3} \dfrac{x^2-9}{x-3} = \dfrac{0}{0}$

$= \lim_{x \to 3} \dfrac{\cancel{(x-3)}(x+3)}{\cancel{(x-3)}} = 6$

$\therefore f(3) \neq \lim_{x \to 3} f(x)$ $\therefore f(x)$ is not continuous at $x=3$

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Ex

Find k for which the function

$$f(x) = \begin{cases} \dfrac{x^2-9}{x-3} & x \neq 3 \\ k-4 & x = 3 \end{cases}$$

is continuous at $x = 3$

Sol

$\because$ $f(x)$ is continuous at $x = 3$.

$$\Rightarrow f(3) = \lim_{x \to 3} f(x)$$

$$k - 4 = \lim_{x \to 3} \frac{x^2-9}{x-3} = 6$$

$$\therefore k - 4 = 6 \Rightarrow k = 10$$

$k-4 = \boxed{262\,554}\ 3 \quad x-3$

$\therefore\ k-4=6 \Rightarrow k=10$

$\boxed{ex}$

If $f(x) = \begin{cases} x^2 & x<1 \\ 5 & x=1 \\ 3x+1 & x>1 \end{cases}$

is $f(x)$ continuous at $x=1$??

sol

① $\therefore f(1) = 5$ ✓

② $\lim\limits_{x\to 1^+} f(x) = \lim\limits_{x\to 1} 3x+1 = 4$

$\lim\limits_{x\to} f(x) - \lim x^2 = 1$

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① $\therefore f(1) = 5$

② $\lim_{x \to 1^+} f(x) = \lim_{x \to 1} 3x+1 = 4$

$\lim_{x \to 1^-} f(x) = \lim_{x \to 1} x^2 = 1$

$\therefore \lim_{x \to 1^+} f(x) \neq \lim_{x \to 1^-} f(x) \neq f(1)$

$\therefore f(x)$ is not continuous.

Dis- ... c- ti...it.. of f

$\therefore \lim_{x \to 1^+} = f(1) \neq \lim_{x \to 1^-}$

$\therefore f(x)$ is not continuous.

* Discuss the continuity of f

$$f(x) = \begin{cases} \dfrac{1-\cos x}{x^2} & x \neq 0 \\ 2 & x = 0 \end{cases}$$

$$f(x) = \begin{cases} \dfrac{\sin 7x}{\tan 3x} & x \neq 0 \end{cases}$$

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$$f(x) = \begin{cases} \dfrac{1-\cos x}{x^2} & x \neq 0 \\ 2 & x = 0 \end{cases}$$

$$f(x) = \begin{cases} \dfrac{\sin 7x}{\tan 3x} & x \neq 0 \\ \dfrac{3}{7} & x = 0 \end{cases}$$

discuss the continuity of $f(x)$ at $x=1$

$$f(x) = \begin{cases} 2x+1 & x \leq 1 \\ x-3 & x > 1 \end{cases}$$

$f(1) = 2(1)+1 = 3$

$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} 2x+1 = 1+1 = 2$

$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} x-3 = 1-3 = -2$

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$\frac{3}{7}$ $x = 0$

discuss the continuity of $f(x)$ at $x = 1$

$$f(x) = \begin{cases} 2x+1 & x \le 1 \\ x^2 - 5 & x > 1 \end{cases}$$

$$f(1) = 2(1)+1 = 3$$

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1} x^2 - 5 = 1 - 5 = -4$$

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1} 2x+1 = 3$$

$$= \lim_{x \to 1^+} \neq \lim_{x \to 1^-}$$ النهاية غير موجودة

$\therefore$ the function $f(x)$ is not continuous at $x = 1$